Pranav Gupta

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Summary

Embedded Systems Engineer with hands-on experience in **STM32/ESP32 firmware, PCB design, motor control**, and **hardware bring-up**. Actively developing metal 3D printer electronics and **real-time control systems** using **Klipper/Marlin, FreeRTOS**, and **Embedded Linux**. Interested in building reliable embedded products for robotics, manufacturing, and industrial systems.

Work Experience

- **ThinkMetal Pvt. Ltd., Chennai, India** | *Embedded Engineer* — Website | *Jul 2025 – Present*
  - Leading **firmware** and **electronics** development for **metal 3D printers** using **STM32**-based control boards, **multi-axis motion systems**, and safety subsystems, improving overall **system stability** across production builds.
  - Customizing **Klipper/Marlin firmware** in **C++** and **Python** for motor control, **thermal regulation**, and **fault handling**, **reducing print failures**.
  - Designing and debugging custom PCBs for **sensor interfaces**, **power regulation**, and **UART/SPI/I2C** communication, improving **peripheral reliability** by **20%**.
  - Improving print **reliability** and **quality** by up to **30%** through **hardware bring-up**, **motion tuning**, and **rapid firmware–hardware iteration**.
  - Ensuring industrial-grade electrical and mechanical **safety** through grounding, **ESD compliance**, **interlocks**, **E-stop** integration, and **SOPs**, reducing electrical and handling risks by **40%**.
- **ThinkMetal Pvt. Ltd., Chennai, India** | *Embedded Engineer Intern* — Website | *Jan 2025 – Jun 2025*
  - Supported development and testing of **embedded firmware** for **metal 3D printer** platforms with **real-time control systems**, **accelerating prototype validation** cycles.
  - Assisted in **PCB design**, **peripheral interfacing**, and **hardware validation** for multi-board embedded systems, reducing integration issues by **15%**.
  - Performed system-level **testing**, **calibration**, and **documentation** to support transition to **production-ready hardware**, improving handover efficiency by **25%**.

Education

- **University Institute of Engineering and Technology, Panjab University** — *2021 – 2025*
  Bachelor of Engineering *in IT* — Chandigarh, India
  - **Relevant Coursework:** Microprocessors, **Digital Electronics**, Computer Networks, IoT, **Embedded C**, Data Structures.
  - **Focus Area:** Embedded systems, **firmware development**, and electronics-oriented problem solving.
  - **Additional Projects:** Line Following Car, Faraday Station, 3D Printer, Secure Archive System (Indian Army).
  - **Major Project:** Voice Assistant — system design, signal processing (DSP) pipeline, and software integration.
  - **Activities & Leadership: Speaker** on Linux fundamentals at **Software Freedom Day '23**, Active member of **Robotics & Programming** Club, Organise Department Visits, Mentoring students, Class Representative.

Skills

- **Embedded Firmware:** Embedded C, C++, STM32CubeIDE, FreeRTOS, Bootloader Basics, Embedded Linux, Klipper.
- **Microcontrollers & Boards:** STM32 (H/F Series, ARM Cortex-M), ESP32, 3D Printer Controller Boards, Raspberry Pi.
- **Communication Protocols:** UART, SPI, I2C, CAN, Ethernet.
- **Electronics & PCB Design:** Schematic Capture, Multi-layer PCB Layout, Altium Designer, KiCad, Power Regulation (Buck & Boost), Motor Drivers, EMI/ESD Practices, Sensors, ADCs.
- **Debugging & Bring-up:** SWD/JTAG Debugging, Multimeter, Oscilloscope, Logic Analyzer, STM32CubeProgrammer.
- **Systems & Software:** Python, Bash, Git, Docker, Embedded Linux Workflows.
- **Professional Skills:** Technical Documentation, System Thinking, Problem Solving, Ownership, Collaboration.

Projects

- **Nano Navigator** [***STM32***, *Embedded C, Motor Control, Sensors*] — *2024*
  - Designed an STM32-based autonomous micromouse system with real-time motor control and sensor fusion.
  - Performed hardware bring-up, power regulation design, and system-level debugging.
- **Bionic Prosthetic Arm** [***Arduino***, ***ESP32***, *EMG/EEG, Servo Control*] — *2022*
  - Developed a bio-signal–driven prosthetic arm integrating EMG/EEG acquisition and embedded control logic.
  - Implemented actuator control and mechanical integration using 3D-scanned anatomical models.
- **Effluent Monitoring System** [***ESP32***, *Sensors, Wi-Fi / BLE*] — *2023*
  - Built an ESP32-based embedded system for real-time effluent level monitoring in industrial tankers.
  - Integrated sensors and wireless communication for live alerts and field-deployable operation.

Achievements / Volunteers

- **1st Place Cognizance (National Level - IIT Roorkee):** E-Conveyor, Faraday Station — *2023, 2024*
- **Mentor @ Summer Internship, DIC, Panjab University:** Embedded Systems, Prosthetics (dicpu.in) — *2022, 2023*
- **President:** Jugaad - Robotics Club, UIET (jugaadclub.in) — *2022, 2023*
- **AIR-7 Technothlon (IIT Guwahati):** National level competition — *2018*